Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
2	(a)	(i)		<u>{organic **/** nitrogenous}</u> base **and** {pentose / 5C sugar} **and** {phosphate / PO_4^{3-} / Pi }	1			1		
		(ii)		Structural Difference ATP has 3 phosphate, RNA has one / ATP has more phosphates / Reference to different bases in ATP and RNA nucleotide or example of (1) RNA has ribose, DNA has dexoxyribose / Less oxygen in DNA nucleotide or description of} / ORA / RNA has {uracil/H} **and** DNA has {thymine/CH_3} (1)	2			2		
		(iii)		{Phosphate/ PO_4^{3-}/ Pi} and {nitrate/NO_3^-}	1			1		
	(b)	(i)		76.9% (2) If incorrect award 1 mark for: 76.923 2 $\times 10^{-11}$ /2.6 $\times 10^{-11}$ x100 (1)		2		2	2	
		(ii)		**Any one (x1) from** Reference to: {protein synthesis / transcription / translation} is taking place / or description of (1) RNA is temporary (1) cell may have more or less ribosomes / OWTTE (1)		1		1		
	(c)			Energy is released when ATP is {hydrolysed / broken down} (to ADP and Pi) / or description of (1) {Energy is stored / ATP is made} when ADP and {phosphate / Pi} are joined / or description of / ADP + Pi → ATP (1) It is the source of energy in all {cells / reactions} in all organisms (1)	1	2		3		
				Question 2 total	**5**	**5**	**0**	**10**	**2**	**0**